MIL-B-44275
3 June 1987
SUPERSEDING
N-B-35K
August 8, 1984

MILITARY SPECIFICATION

BAKERY MIXES, PREPARED

This specification is approved for use in all Departments and agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This document covers some of the types and classes of commercially available prepared bakery mixes that require only the addition of water for preparation and baking, for use by the Department of Defense. The type II mixes are specially prepared for extended shelf life and use at all altitudes by the Department of Defense as items of general issue.

1.2 Classification.

1.2.1 Types, classes, and styles. Prepared bakery mixes shall be of the following types, classes, and styles, as specified (see 6.1).

Type I - Commercially available mixes
Type II - Extended shelf life mixes

Class 1 - Cake mixes

Style A - White
Style B - Yellow
Style C - Devil's food

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Style D - Gingerbread
Style E - Pound (type I only)
Style F - Angel food (type I only)
Style G - Banana (type I only)
Style H - Spice (type I only)
Style I - Orange (type I only)
Style J - Lemon (type I only)
Style K - Carrot (type I only)
Style L - German chocolate (type I only)

Class 2 - Quick bread mixes

Style A - Yellow corn bread
Style B - Biscuit mix
Style C - Pancake and waffle, regular
Style D - Pancake and waffle, buttermilk (type I only)
Style E - Pancake, buckwheat (type I only)

Class 3 - Cake doughnut mix

Class 4 - Yeast-leavened mixes

Style A - Roll mix
Style B - Sweet dough

Class 5 - Brownie mix

Style A - Chocolate
Style B - Butterscotch

Class 6 - Muffin mixes

Style A - Plain (type I only)
Style B - Bran (type I only)

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

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SPECIFICATIONS

FEDERAL

UU-S-48	-	Sacks, Shipping, Paper
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing of

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-10547	-	Liners, Case and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-L-35078	-	Loads, Unit: ° Preparation of Semiperishable Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting activity.)

2.1.2 Other government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the day of the solicitation.

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder (21 CFR PARTS 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing the Inspection of Eggs and Egg Products

(Application for copies should be addressed to the Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

United States Standards for Grades of Dry Sweetcream Buttermilk

United States Standards for Grades of Nonfat Dry Milk
(Spray Process)

United States Standards for Dry Whey

(Application for copies should be addressed to Director, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association
of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

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AMERICAN ASSOCIATION OF CEREAL CHEMISTS

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DRY MILK INSTITUTE, INC.

Standards for Grades of Dry Milks, Including Methods of Analysis, Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 North Franklin Street, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1120, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Ingredients (type II only). In addition to the requirements of 3.2, the following ingredients for type II mixes shall meet the specific requirements of 3.2.1.1 through 3.2.1.32.

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3.2.1.1 Wheat flour. Wheat flour shall be enriched flour of the nominal type specified for each product and shall be of such quality as to provide the required characteristics of the end item. Flour may be brominated according to FDA guidelines when used in muffin mixes.

3.2.1.2 Sugar. Sugar shall be a white, refined cane or beet sugar, or a combination thereof. The sugar shall be of a particle size known as "Baker's Special" which passes through a U.S. Standard No. 40 sieve, but not more than 20.0 percent by weight, passes through a U.S. Standard No. 140 sieve.

3.2.1.3 Dried egg products. Dried egg products shall be dried egg yolks, dried egg whites, or dried whole eggs as applicable to produce the required product. The dried egg products shall be processed and labeled in accordance with the Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59) as evidence by the USDA egg products inspection shield on the label. Dried egg products shall not exceed 6 months of age at the time of their incorporation.

3.2.1.4 Shortening. The shortening shall be refined hydrogenated vegetable oil of 100 hours or greater stability as determined by the Active Oxygen Method (AOM). In addition, the shortening shall contain FDA approved antioxidants at levels not to exceed regulatory guidelines. The free fatty acids shall not exceed 0.08 percent. If an emulsified shortening is used, it shall comply with the above AOM stability but the free fatty acids shall not exceed 0.15 percent.

3.2.1.5 Salt. Salt shall be white refined sodium chloride with or without anticaking agent. Iodized salt shall not be used.

3.2.1.6 Milk, nonfat dry (high heat). Nonfat dry milk shall be U.S. Extra Grade, High Heat as defined in the U.S. Standards for Grades of Nonfat Dry Milk (Spray Process).

3.2.1.7 Chemical leavening agents. Chemical leavening agents shall be Food Chemicals Codex purity. Permitted ingredients shall be sodium bicarbonate, sodium aluminum phosphate, and coated monocalcium phosphate. The coating shall be a hard fat or other suitable material which will slow the solubility of the salt. The above ingredients shall be powdered and free flowing.

3.2.1.8 Mono- and diglycerides. Mono- and diglycerides shall be a combination of monoglycerides and diglycerides. The manufacturer's description of the product shall indicate an alpha monoglyceride content in a range between 40 to 55 percent.

3.2.1.9 Monoglycerides. Monoglycerides shall be essentially pure distilled monoglycerides, either hard or plastic as specified.

3.2.1.10 Sodium stearoyl-2-lactylate and sodium or calcium propionate. Sodium stearoyl-2-lactylate and sodium or calcium propionate shall be powdered and free flowing and comply with the requirements of Food Chemicals Codex.

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3.2.1.11 Cocoa. Cocoa shall be medium fat Dutch process cocoa of the commercial type known as "medium Dutched". Neither sodium carbonate nor sodium bicarbonate shall be used in the Dutching (alkali treatment) process. Malaysian beans shall not be used. It shall conform to the Definition and Standards for cocoa products under the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.2.1.12 Dried molasses. Dried molasses shall be a free flowing product made from dark molasses and suitable carrying material such as sugar, cornstarch, flour and other carrying ingredients. Also, it may contain free flow agents such as silicon dioxide. Molasses solids content shall be between 62 to 66 percent.

3.2.1.13 Caramel color. Caramel color shall be full strength powdered caramel color of National Formulary grade.

3.2.1.14 Buckwheat flour. Buckwheat flour shall be free flowing.

3.2.1.15 Yellow corn meal. Yellow corn meal shall be degermed yellow corn, of a granulation which will impart the characteristic grittiness in corn bread.

3.2.1.16 Yellow corn flour. Yellow corn flour shall be finely powdered yellow corn resembling cornstarch, except for color. It shall not impart grittiness to the baked products in which it is used.

3.2.1.17 Buttermilk, sweetcream, dry. Dry sweetcream buttermilk shall be U.S. Extra Grade as defined in the U.S. Standards for Grades of Dry Sweetcream Buttermilk.

3.2.1.18 Corn syrup solids. Corn syrup solids shall be of the commercial type known as low dextrose equivalent. The manufacturer's description of the product shall indicate a dextrose equivalent in a range from 15 to 30.

3.2.1.19 Brown sugar. Brown sugar shall be of the commercial types known as light brown or medium brown sugar.

3.2.1.20 Wheat starch. Wheat starch shall be white refined starch made from wheat flour. Chemical analysis for wheat starch shall be as follows:

Percent by weight (12.5 percent moisture basis)

Ash not more than 0.25
Fat not more than 0.52

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3.2.1.21 Soy albumen. Soy albumen shall be derived from the enzymatic treatment of soy. Chemical analysis of this product shall be as follows:

	Percent by weight
Protein (N x 6.25)	not less than 65.0
Moisture	not more than 3.5

3.2.1.22 Dried egg whites with whipping aid. The dried egg whites shall meet the requirements of 3.2.1.3 and contain not more than 0.1 percent of sodium lauryl sulfate as a whipping aid. The dried egg white shall be the type(s) used in white and angel food cake mixes.

3.2.1.23 Low ash cake flour. Low ash cake flour shall be milled specifically for angel food cake production. The extraction shall be such that the ash content shall be not more than 0.33 percent by weight. The particle size shall be not greater than 20 microns average. The fat content shall be not more than 0.6 percent by weight. Requirements shall be on a 12.5 percent moisture basis.

3.2.1.24 Sodium silicoaluminate. The sodium silicoaluminate shall comply with the requirements of Food Chemicals Codex.

3.2.1.25 Soluble spice flavoring. Soluble spice flavoring shall be the flavoring constituents of the natural spices incorporated in a soluble dry carrier, such as corn syrup solids or dextrose, in such a ratio that the spice flavoring is equivalent to an equal weight of natural spice.

3.2.1.26 Soy flour (defatted). Soy flour (defatted) shall be the finely milled free flowing powder obtained from the processing of soybeans.

3.2.1.27 Anhydrous monocalcium phosphate. Anhydrous monoclacium phosphate shall meet the requirements of Food Chemicals Codex.

3.2.1.28 Lecithin. Lecithin shall be a concentrate manufactured from soybeans and shall comply with the Food Chemicals Codex.

3.2.1.29 Dextrose. Dextrose shall be a white crystalline sugar commercially obtained by complete hydrolysis of starch. It shall be powdered and free-flowing.

3.2.1.30 Lecithinated soy flour. Soy flour (3 to 6 percent lecithin) shall be made from soy flour with a 70 PDI (protein dispersibility index) and should have a minimum protein content of 50 percent and contain 4.5 $\pm$ 1.5 percent lecithin.

3.2.1.31 Whey, dry (sweet-type). Dry sweet-type whey shall be U.S. Extra Grade, as defined in the U.S. Standards for Dry Whey.

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3.2.1.32 Flavor, butterscotch. Butterscotch flavor shall be a natural or synthetic spray-dried flavor.

3.3 Preparation mix and prepared product requirements.

3.3.1 Type I. Commercially available mixes.

3.3.1.1 Class 1, cake mixes. Styles A through E and styles G through L cake mixes shall be prepared from sugar, bleached soft wheat flour, shortening, nonfat dry milk, emulsifier, salt, leavening agents, the ingredients specified for the styles in 3.3.1.1.1 through 3.3.1.1.5 and 3.3.1.1.7 through 3.3.1.1.12, and other ingredients which will produce the required product. Style F shall be prepared from the ingredients specified in 3.3.1.1.6. The finished cake mix shall not contain more than 6.0 percent moisture, by weight, except the gingerbread mix shall not contain more than 7.0 percent moisture, by weight, the pound cake mix shall not contain more than 5.5 percent moisture, by weight, the carrot cake mix and the German chocolate cake mix shall not contain more than 4.8 percent moisture, by weight, and the albumen mix in the angel food cake mix, if packaged separately, shall not contain more then 3.0 percent moisture, by weight. When the finished cake mixes are prepared according to 4.6.1.1, they shall produce a moist, close-grained, smooth textured product having a characteristic odor, flavor, and color for its style and having specific volume, when examined in accordance with 4.6.2, as follows:

Cake Mixes

Cake	Specific volume not less than
White	3.0
Yellow	3.0
Devil's food	3.0
Gingerbread	2.0
Pound	1.9
Angel food	4.5
Banana	2.5
Spice	3.0
Orange	2.5
Lemon	2.5
Carrot	2.4
German chocolate	3.0

All styles, except styles E and F, shall not have breaks or cracks when examined in accordance with 4.6.3.

3.3.1.1.1 Style A, white cake mix. In addition to the ingredients specified in 3.3.1.1, white cake mix shall contain dried egg whites and flavor.

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3.3.1.1.2 Style B, yellow cake mix. In addition to the ingredients specified in 3.3.1.1, yellow cake mix shall contain dried whole eggs or dried egg yolks and dried egg whites and flavor.

3.3.1.1.3 Style C, devil's food cake mix. In addition to the ingredients specified in 3.3.1.1, devil's food cake mix shall contain cocoa, dried whole eggs or dried egg yolks and dried egg whites, and flavor.

3.3.1.1.4 Style D, gingerbread cake mix. In addition to the ingredients specified in 3.3.1.1, gingerbread cake mix shall contain molasses solids, caramel color, and ground spices or spice flavorings.

3.3.1.1.5 Style E, pound cake mix. In addition to the ingredients specified in 3.3.1.1, pound cake mix shall contain dried whole eggs.

3.3.1.1.6 Style F, angel food cake. Angel food cake mix shall be prepared from sugar, wheat flour, dried egg whites, salt, flavors, and other ingredients which will produce the required product. Dried egg whites, with other ingredients as necessary, may be packaged separately from the basic mix.

3.3.1.1.7 Style G, banana cake mix. In addition to the ingredients specified in 3.3.1.1, banana cake mix shall contain dried whole eggs or dried egg yolks and dried egg whites, and dried bananas.

3.3.1.1.8 Style H, spice cake mix. In addition to the ingredients specified in 3.3.1.1, spice cake mix shall contain dried whole eggs or dried egg yolks and dried egg whites and ground spices or spice flavoring.

3.3.1.1.9 Style I, orange cake mix. In addition to the ingredients specified in 3.3.1.1, orange cake mix shall contain dried whole eggs or dried egg yolks and dried egg whites and orange flavoring.

3.3.1.1.10 Style J, lemon cake mix. In addition to the ingredients specified in 3.3.1.1, lemon cake mix shall contain dried whole eggs or dried egg yolks and dried egg whites and lemon flavoring.

3.3.1.1.11 Style K, carrot cake mix. In addition to the ingredients specified in 3.3.1.1, carrot cake mix shall contain raisins, almond pieces, coconut, whey, dried carrots, egg yolk, modified cornstarch, corn syrup solids, artificial and natural flavors, xanthan gum, sodium caseinate, artificial food colors, mono- and diglycerides, butylated hydroxyanisole, and sodium sulfite.

3.3.1.1.12 Style L, German chocolate cake mix. In addition to the ingredients specified in 3.3.1.1, German chocolate cake mix shall contain cocoa (medium Dutched), dextrose, egg white, egg yolk, corn syrup solids, artificial flavors, mono- and diglycerides, xanthan gum and sodium caseinate.

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3.3.1.2 Class 2, quick bread mixes. Quick bread mixes shall be prepared from wheat flour, sugar, shortening, nonfat dry milk, salt, leavening agents, and other ingredients which will produce the required product as specified in 3.3.1.2.1 through 3.3.1.2.5. The products when prepared in accordance with 4.6.1.2 through 4.6.1.4 shall not have a pronounced chemical leavening flavor.

3.3.1.2.1 Style A, yellow corn bread mix. Yellow corn bread mix shall comply with 3.3.1.2. In addition to the ingredients specified in 3.3.1.2, the finished mix shall contain yellow degermed corn meal and dried whole eggs or dried egg yolks and dried egg whites. The moisture content shall be not more than 10.0 percent by weight. The baked product when prepared according to 4.6.1.2 shall have a specific volume of not less than 2.2 when examined in accordance with 4.6.2. The upper surface shall be slightly convex, without dips and large cracks, and the crust shall be smooth and have a uniform golden brown color. The baked product shall withstand handling without breaking or crumbling, but shall be sufficiently tender to be readily broken into large pieces by hand. The crumb shall have a uniform grain and be free from lumps or pieces of unincorporated mix.

3.3.1.2.2 Style B, biscuit mix. The biscuit mix shall comply with 3.3.1.2. The finished mix shall have a moisture content by weight of not more than 10.0 percent. The baked product when prepared in accordance with 4.6.1 and 4.6.1.3 shall have a specific volume of not less than 2.4 when examined in accordance with 4.6.2. The biscuits shall be symmetrical, the crust shall have a substantially uniform light brown color, the top and bottom portions shall be easily separated by hand, and the crumb shall be tender, white, and composed primarily of small, thin-walled cells and shall not be soggy or gritty.

3.3.1.2.3 Style C, pancake and waffle mix, regular. Regular pancake and waffle mix shall comply with 3.3.1.2 except that dextrose may be substituted for a portion of the sugar, and shortening may or may not be used. In addition, the product shall contain corn flour, dried whole eggs or dried egg yolks and dried egg whites, and other ingredients which will produce the required mix. The leavening agents shall be sodium aluminum phosphate or sodium aluminum sulfate. The moisture content of the mix shall be not more than 10.0 percent by weight. The fried or griddled product when prepared in accordance with 4.6.1.4 shall conform to the following requirements: the height at the center shall be not less than 1/4 inch nor more than 5/8 inch: the crust shall be light brown; and the crumb shall be light yellow and small celled, and shall not be tough or doughy.

3.3.1.2.4 Style D, pancake and waffle mix, buttermilk (yeast not included due to its limited shelf life). Buttermilk pancake and waffle mix shall conform to the requirements of 3.3.1.2.3 and in addition shall contain added buttermilk solids.

3.3.1.2.5 Style E, pancake, buckwheat mix. Buckwheat pancake mix shall conform to the requirements of 3.3.1.2.3 except that buckwheat flour shall be used in lieu of corn flour and the crumb color shall be light brown.

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3.3.1.3 Class 3, cake doughnut mix. Cake doughnut mix shall be prepared from wheat flour, sugar, shortening, nonfat dry milk, dried whole eggs or dried egg yolks and dried egg whites or a combination thereof, salt, leavening agents, and other ingredients which will produce the required product. The finished mix shall contain not more than 8.5 percent moisture by weight. The fried product when prepared in accordance with 4.6.1.5 shall conform to the following requirements: the specific volume shall be not less than 2.6 when examined in accordance with 4.6.2. The fried product shall have a uniform brown color and shall be substantially symmetrical without excessive bulges, blisters or cracks. The inner hole shall be puckered and small. The grain shall be fine and uniform, and the depth of grease penetration shall not exceed 1/16 inch. The manufacturer's instructions shall include water requirements for both a hand-cut and a machine-cut doughnut.

3.3.1.4 Class 4, yeast-leavened mixes. Yeast-leavened mixes shall be prepared from wheat flour, sugar, shortening, high heat nonfat dry milk, salt and other ingredients which will produce the required product. The proper amount of active dry yeast shall be unit packed separately (see 5.1) and placed on top of the product in the basic container prior to final sealing. The moisture content of the finished mixes shall be not more than 12.0 percent by weight.

3.3.1.4.1 Style A, roll mix. The mix shall be baked into a loaf for testing. The baked loaf when prepared in accordance with 4.6.1 and 4.6.1.6 shall conform to the following requirements: the loaf volume shall be not less than 2400 milliliters when examined in accordance with 4.6.2; the loaf shall be smooth and without breaks. The grain shall be fine and uniform, and the air cells shall be slightly elliptical with thin walls. The crumb shall be free from large holes and shall be smooth, soft, and velvety.

3.3.1.4.2 Style B, sweet dough mix. Sweet dough mix shall comply with 3.3.1.4 and in addition, shall contain dried whole eggs or dried egg yolks and dried egg whites or combinations thereof. The baked loaf when prepared in accordance with 4.6.1.7 shall conform to the baked product requirements of 3.3.1.4.1 except that the loaf volume shall be not less than 2050 milliliters when examined in accordance with 4.6.2.

3.3.1.5 Class 5 brownie mixes.

3.3.1.5.1 Style A, chocolate brownie mix. Chocolate brownie mix shall be prepared from sugar, wheat flour, shortening, cocoa, dried whole eggs or dried egg yolks and dried egg whites or a combination thereof, salt, leavening agents, flavor, and other ingredients which will produce the required product. The finished product mix shall contain not more than 3.5 percent moisture by weight. The baked product shall conform to the following requirements: when prepared in accordance with 4.6.1.8, the specific volume shall be not less than 1.5 when examined in accordance with 4.6.2. The product shall be sufficiently tender to be sliced easily and shall not be soggy, rubbery, or excessively hard.

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3.3.1.5.2 Style B, butterscotch brownie mix. Butterscotch brownie mix shall be prepared from sugar, wheat flour, shortening, dried whole egg solids or dried egg yolk and egg whites, salt, leavening agents, wheat starch, caramel and butterscotch flavors, and other ingredients which will produce the required product. When baked according to 4.6.1.8, the specific volume shall be not less than 1.5 when examined in accordance with 4.6.2. The product shall be chewy but not sticky or gummy and shall have a pleasing butterscotch-type flavor.

3.3.1.6 Class 6, muffin mixes.

3.3.1.6.1 Style A, plain mix. Style A plain muffin mix shall be prepared from either enriched bleached flour or enriched brominated flour (niacin, iron, thiamine mononitrate, riboflavin), sugar hydrogenated vegetable oil (cottonseed, soybean, palm, or combinations), dextrose, whey, salt, sodium acid pyrophosphate or sodium aluminum phosphate, baking soda, egg yolks or egg whites or combinations of both, nonfat milk, lecithin, artificial flavor, monocalcium phosphate. Optional ingredients may be added as potassium bromate, malted barley flour, xanthan gum, dried corn syrup, natural colors (annatto, turmeric), sodium caseinate. The finished product mix shall contain not more than 9.0 percent moisture by weight. When baked in a pan (2 ounce units, #20 dipper), according to commercial directions, the muffins shall peak to a smooth convex point without forming "noses", possess a firm browned crust with a uniform internal structure free of tunnels. The product shall be free of gumminess and shall possess the flavor of a typical baked plain muffin, free of off flavors and odors.

3.3.1.6.2 Style B, bran mix. Style B bran muffin mix shall be prepared from either enriched bleached flour or enriched brominated flour (niacin, iron, thiamine nomonitrate, riboflavin), sugar, wheat bran, hydrogenated vegetable oil (cottonseed, soybean, palm, or combinations), whey, dried molasses or dried honey, baking soda, salt, egg whites or egg yolks, sodium acid pyrophosphate or sodium aluminum phosphate, or soy flour. Optional ingredients may be added as dextrose, sodium caseinate, maltodextrin, wheat starch, monocalcium phosphate. The finished product mix shall contain not more than 9.0 percent moisture by weight. When baked in a pan (2 ounce units, #20 dipper), according to commercial directions, the muffins shall peak to a smooth convex point without forming "noses", possess a firm browned crust with a uniform internal structure free of tunnels. The product shall be free of gummines and shall possess the flavor of a typical baked bran muffin, free of off flavors and odors.

3.3.2 Type II. Extended shelf life mixes. For the type II mixes, sodium silicoaluminate may be substituted for up to 1.0 percent by weight of the formulation for equal parts of flour and sugar for class 1, styles A, B, C and D cake mixes, and class 5, styles A and B brownie mixes, and for an equal amount of flour for class 2, styles A and B quick bread mixes.

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3.3.2.1 Class 1, cake mixes. Styles A, B, and C, products shall have a moisture content of not more than 5.5 percent by weight. Style D product shall have a moisture content of not more than 7.0 percent by weight. Bicarbonate of soda, when required, shall be packaged separately as specified in 5.1. The individual cake shall have a pleasing flavor and aroma typical of the product as formulated.

3.3.2.1.1 Style A, white cake mix. The baked product, when prepared according to 4.6.1.1, shall have a golden brown crust with a slightly convex top surface, free of cracks. The crumb shall be characteristically moist, white and evenly fine textured approaching the texture of an angel food cake. When bisected horizontally with a sharp saw-toothed knife, the cake shall have no large gas pockets or tunnels. Two hours after baking, the entire cake shall separate easily from the pan by inverting the pan and tapping one edge lightly on a hard surface. The cake shall not break or crack when examined in accordance with 4.6.3. The cake shall have a specific volume of 2.9 or greater immediately after cooling when examined in accordance with 4.6.2. The formulation for white cake shall be as follows:

White cake mix 1/ 4/

Ingredients	Percent (weight)
Sugar	43.00
Shortening	9.00
Salt	0.80
Mono- and diglycerides 2/ 3/	0.90
Sodium aluminum phosphate	0.50
Sodium stearoyl-2-lactylate 3/	0.40
Wheat flour, cake type	40.80
Lecithin 3/	0.10
Egg whites, modified	2.00
Soy flour (3 to 6 percent lecithin)	1.00
Sweet dairy whey	1.00
Bicarbonate of soda (pouch, see 5.1)	0.50

1/ The batter shall be able to withstand a 30-minute setting period in a round pan, at a temperature of 78° to 84°F without exhibiting a crusty foam-like top after baking.

2/ Mono- and diglycerides shall be incorporated into the shortening by the shortening manufacturer.

3/ An emulsifier system which produces equivalent results may be substituted, provided the resultant cake meets all other requirements of this specification. Lecithin must be used in either case.

4/ Mixing of ingredients shall be by methods which insure uniform distribution. Use of a turbulizer (impact mill) or equivalent equipment is recommended.

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3.3.2.1.2 Style B, yellow cake mix. The baked product when prepared according to 4.6.1.1, shall produce a cake with a golden brown crust having a slightly convex top surface free of cracks. The crumb shall be characteristically moist, light yellow and slightly open textured. When bisected horizontally with a sharp saw-toothed knife, the cake shall have no large gas pockets or tunnels. Two hours after baking, the entire cake shall separate easily from the pan by inverting the pan and tapping one edge lightly on a hard surface. The cake shall not break or crack when examined in accordance with 4.6.3. The cake shall have a specific volume of 3.0 or greater immediately after cooling when examined according to 4.6.2. The formulation for yellow cake shall be as follows:

Yellow cake mix 4/ 5/

Ingredients 1/	Percent (weight)
Sugar	40.90
Shortening	8.70
Sodium aluminum phosphate	0.55
Salt	0.90
Sodium Stearoyl-2-lactylate 3/	0.40
Mono- and diglycerides 2/ 3/	0.90
Wheat flour, cake type	40.00
Lecithin 3/	0.10
Dried whole eggs	5.00
Nonfat dried milk	2.00
Bicarbonate of soda (pouch, see 5.1)	0.55

1/ To each 1,000 pounds of mix add 16.75 grams of FD&C yellow No. 5.

2/ Mono- and diglycerides shall be incorporated into the shortening by the shortening manufacturer.

3/ An emulsifier system which produces equivalent results may be substituted provided the resultant cake meets all other requirements of this specification. Lecithin must be used in either case.

4/ The batter shall be able to withstand a 30-minute setting period in a round pan, at a temperature of 78° to 84°F without exhibiting a crusty, foam-like top after baking.

5/ Mixing of ingredients shall be by methods which insure uniform distribution. Use of a turbulizer (impact mill) or equivalent equipment is recommended.

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3.3.2.1.3 Style C, devil's food cake mix. The product, when prepared according to 4.6.1.1, shall have a slightly open texture, a reddish to brownish black slightly convex crust free of cracks, and a reddish-black moist crumb. When bisected horizontally with a sharp saw-toothed knife, the cake shall have no large gas pockets or tunnels. Two hours after baking, the entire cake shall separate easily from the pan by inverting the pan and tapping one edge lightly on a hard surface. The cake shall not break or crack when examined in accordance with 4.6.3. The cake shall have a specific volume of 2.7 or greater immediately after cooling when examined in accordance with 4.6.2. The pH of the baked product shall be 9.0 or greater. The formulation for devil's food cake mix shall be as follows:

Devil's food cake mix 3/ 4/

Ingredients	Percent (weight)
Sugar	41.80
Shortening	9.00
Polysorbate 60 1/ 2/	0.10
Sorbitan monostearate 1/ 2/	0.40
Hard monoglycerides 1/ 2/	0.25
Plastic monoglycerides 1/ 2/	0.25
Salt	0.75
Wheat flour, cake type 5/	36.50
Lecithin 2/	0.10
Cocoa (Dutch process)	4.00
Nonfat dried milk	0.70
Dried whole eggs	3.00
Soy flour (3 to 6 percent lecithin)	2.00
Bicarbonate of soda (pouch, see 5.1)	1.15

1/ The emulsifiers shall be incorporated into the shortening by the shortening manufacturer.

2/ An emulsifier system which produces equivalent results may be substituted provided the resultant cake meets all other requirements of this specification. Lecithin must be used in either case.

3/ The batter shall be able to withstand a 30-minute setting period in a round pan, at a temperature of 78° to 84°F without exhibiting a crusty, foam-like top after baking.

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4/ Mixing of ingredients shall be by methods which insure uniform distribution. Use of a turbulizer (impact mill) or equivalent equipment is recommended.

5/ Not to exceed 13.5 percent moisture.

3.3.2.1.4 Style D, gingerbread cake mix. The product, when prepared according to 4.6.1.1, shall produce a cake with a brown crust having a slightly convex top surface free from cracks. The crumb shall be characteristically moist, brownish yellow in color, and of a texture midway between a yellow cake texture and a chewy high gluten, low specific volume bread. When bisected horizontally with a sharp saw-toothed knife, the gingerbread shall have no large gas pockets or tunnels. Two hours after baking, the entire cake shall separate easily from the pan by inverting the pan and tapping one edge slightly on a hard surface. The cake shall not break or crack when examined in accordance with 4.6.3. The gingerbread shall have a specific volume of 2.2 or greater immediately after cooling when examined according to 4.6.2. The pH of the baked gingerbread shall be not less than 8.0. The formulation for gingerbread cake mix shall be as follows:

Gingerbread cake mix 2/

Ingredients	Percent (weight)
Sugar	30.50
Shortening	9.00
Dried molasses	10.00
Salt	0.70
Wheat flour	45.40
Caramel color	0.10
Spice mix 1/	2.00
Soy flour (3 to 6 percent lecithin)	0.65
Sweet dairy whey	0.65
Bicarbonate of soda (pouch, see 5.1)	1.00

1/ The soluble spice flavoring shall be in a ratio of 5 parts ginger, 4 parts cinnamon and 2 parts allspice by weight.

2/ Mixing of ingredients shall be by methods which insure uniform distribution. Use of a turbulizer (impact mill) or equivalent equipment is recommended.

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3.3.2.2 Class 2, quick bread mixes.

3.3.2.2.1 Style A, yellow corn bread mix. The corn bread, when prepared in accordance with 4.6.1.2, shall be an even grained product with a slightly convex top surface without dips or cracks 1 hour after baking. The specific volume of the baked product shall not be less than 2.0, when examined in accordance with 4.6.2. The moisture content of the dry mix shall be no more than 9.5 percent by weight. The formulation shall be as follows:

Yellow corn bread mix

Ingredients	Percent (weight)
Wheat flour, bread type	41.00
Yellow corn meal	32.00
Shortening	11.00
Sugar	2.50
Nonfat dried milk	5.00
Dried whole eggs	5.00
Salt	1.00
Sodium aluminum phosphate	1.00
Bicarbonate of soda (pouch, see 5.1)	1.00
Sodium stearoyl-2-lactylate	0.50

3.3.2.2.2 Style B, biscuit mix. The biscuit mix, when prepared in accordance with 4.6.1.3, shall be symmetrical and the crust shall have a substantially uniform light brown color. The crumb shall be tender, white, and composed primarily of small, thin walled cells, and shall not be soggy or gritty. The specific volume of the baked product shall be not less than 2.4 when examined in accordance with 4.6.2. The moisture content of the mix shall be not more than 10.0 percent by weight. The formulation for biscuit mix shall be as follows:

Biscuit mix

Ingredients	Percent (weight)
Sugar	2.60
Shortening	15.00
Wheat flour, bread type 1/	73.20
Soy flour (3 to 6 percent lecithin)	1.60
Sweet dairy whey	3.20
Salt	1.70
Sodium aluminum phosphate (pouch, see 5.1)	1.35
Bicarbonate of soda (pouch, see 5.1)	1.35

1/ Not to exceed 13.5 percent moisture.

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3.3.2.2.3 Pancake and waffle mix. Style C pancake and waffle mix shall contain not more than 10.0 percent moisture by weight. Style C, when prepared according to 4.6.1.4, shall conform to the following requirements: The height at the center shall be not less than 3/8 inch nor more than 5/8 inch. The crust shall be light brown. The crumb shall be light yellow, small celled, and shall not be tough, doughy nor harsh flavored. The flour used for the pancake mix may be bleached. The formula for the mix shall be as follows:

Style C pancake and waffle mix, regular

Ingredients	Percent (weight)
Wheat flour, pastry type	60.00
Yellow corn flour	15.50
Sugar	6.00
Dried whole eggs	3.85
Soy flour (3 to 6 percent lecithin)	2.00
Sweet dairy whey	4.15
Shortening	3.00
Sodium aluminum phosphate	2.00
Salt	1.50
Bicarbonate of soda (pouch, see 5.1)	2.00

3.3.2.3 Class 3, cake doughnut mix all-purpose. Cake doughnut mix shall contain not more than 9 percent moisture by weight. The fried product, when prepared according to 4.6.1.5, shall conform to the following requirements: The specific volume shall be not less than 2.4 when examined in accordance with 4.6.2. The product shall have a uniform brown color and shall not have major bulges, blisters, or cracks. The grain shall be fine and uniform without large holes or tunnels and the depth of grease penetration shall not exceed 1/16 inch. The manufacturer's instructions shall include water requirements for hand-cut and machine-cut doughnuts. The formulation for cake doughnut mix shall be as follows:

Cake doughnut mix

Ingredients	Percent (weight)
Doughnut flour 1/	62.90
Sugar	21.00
Shortening	3.25
Dried whole eggs	2.50
Defatted soy flour	2.50
Lecithin	0.15
Anhydrous monocalcium phosphate	0.16
Sodium aluminum phosphate	0.80
Salt	1.00
Nonfat dry milk	3.00

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Cake doughnut mix (cont'd)

Ingredients	Percent (weight)
Dextrose	1.50
Egg whites, modified	0.30
Bicarbonate of soda (pouch, see 5.1)	0.94

1/ Flour shall be a blend of wheat flour, with little or no bleaching, specifically milled for cake doughnuts.

3.3.2.4 Class 4, yeast-leavened mixes.

3.3.2.4.1 Style A, roll mix. The baked loaf, when prepared in accordance with 4.6.1.6, shall conform to the following requirements: the loaf volume shall be not less than 2400 milliliters when examined in accordance with 4.6.2. The loaf shall be smooth and without wide breaks. When the cooled loaf is bisected vertically through the center with a sharp knife, the grain shall be fine and uniform, and the air cells shall be slightly elliptical with thin walls. The crumb shall be free from large holes. Texture shall be smooth, soft, and velvety. Roll mix shall contain not more than 12 percent moisture by weight. The formulation shall be as follows:

Roll mix 1/

Ingredient	Percent (weight)
Wheat flour, bread type	84.00
Sugar	4.20
Shortening	4.25
Soy flour (3 to 6 percent lecithin)	2.00
Sweet dairy whey	2.25
Salt	2.00
Sodium Stearoyl-2-lactylate 2/	0.50
Calcium propionate	0.25
Monodiglycerides 3/ 2/	0.55

1/ Mixing of ingredients shall be by methods which insure uniform distribution. Use of a turbulizer (impact mill) or equivalent equipment is recommended.

2/ An emulsifier system which produces equivalent results may be substituted provided the resultant roll mix meets all other requirements of this specification.

3/ Monodiglycerides shall be incorporated into the shortening by the shortening manufacturer.

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3.3.2.4.2 Style B, sweet dough mix. The baked loaf, when prepared in accordance with 4.6.1.7, shall conform to the baked product requirements of 3.3.2.4.1 except that the loaf volume shall be not less than 2050 milliliters when examined in accordance with 4.6.2. The moisture content of the mix shall be not more than 9.5 percent by weight. The formulation shall be as follows:

Sweet dough mix 1/

Ingredients	Percent (weight)
Wheat flour, bread type 2/	66.50
Sugar	12.00
Shortening	12.00
Nonfat dry milk	3.50
Dried whole eggs	3.25
Salt	1.25
Mono-diglycerides 3/ 4/	1.00
Sodium stearoyl-2-lactylate 4/	0.50

1/ Mixing of ingredients shall be by methods which insure uniform distribution. Use of a turbulizer (impact mill) or equivalent equipment is recommended.

2/ Not to exceed 13.5 percent moisture.

3/ Mono- and diglycerides shall be incorporated into the shortening by the shortening manufacturer.

4/ An emulsifier system which produces equivalent results may be substituted provided the resultant sweet dough mix meets all requirements of this specification.

3.3.2.5 Class 5, brownie mix.

3.3.2.5.1 Style A, chocolate brownie mix. The baked product, when prepared according to 4.6.1.8, shall be chewy, but not gummy, and shall be capable of being cut into squares which are easily handled without crumbling or breaking. The mix shall have a flavor which is unmistakably chocolate, but shall not be harsh or bitter. The moisture content for chocolate brownie mix shall be not more than 4.0 percent by weight. The mix shall be formulated as follows:

Chocolate brownie mix

Ingredients	Percent (weight)
Sugar	47.50
Corn syrup solids	4.00
Wheat flour, bread type	25.00
Egg whites, modified	3.00
Shortening	9.00

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Chocolate brownie mix (cont'd)

Ingredients	Percent (weight)
Cocoa	7.00
Nonfat dry milk	3.00
Salt	0.90
Sodium aluminum phosphate	0.30
Bicarbonate of soda (pouch, see 5.1)	0.30

3.3.2.5.2 Style B, butterscotch brownie mix. The baked butterscotch brownie, when prepared according to 4.6.1.8, shall be chewy, but not gummy and shall be capable of being cut into squares which are easily handled without crumbling or breaking. The product shall have a pleasing butterscotch and brown sugar flavor. The moisture content for butterscotch brownie mix shall be not more than 7.5 percent by weight. Butterscotch brownie mix shall be formulated as follows:

Butterscotch brownie mix

Ingredients	Percent (weight)
Wheat flour, bread type	35.00
Brown sugar	28.00
Granulated sugar	22.00
Shortening	10.00
Egg whites, modified	3.50
Salt	1.00
Sodium aluminum phosphate	0.25
Bicarbonate of soda (pouch, see 5.1)	0.25

3.4 Finished product. The finished product shall be a well-blended free-flowing homogeneous mixture. The mix shall be free from dirt, filth, mustiness, extraneous matter, objectionable odor(s), and lumps other than those which disintegrate readily upon application of slight finger pressure. In addition, each finished bakery mix shall meet the applicable requirements specified in 3.3 for the pertinent product.

3.4.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110: "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

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3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to any identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5.1.2 Pouch material examination. Conformance of pouch material to 5.1 shall be ascertained by examination of pertinent invoices, labels or other valid documents from suppliers of pouches or pouch material. Nonconformance shall indicate unacceptable pouches or pouch material and use of such shall be cause for rejection of the involved end item.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to requirements for mixing, formulations, and packaging. Records of formulations shall be maintained. Any nonconformance revealed by actual examination or by review of records of time and formulation or other valid documents shall be cause for rejection of the involved product.

4.5.3 Net weight inspection. The filled and sealed primary containers shall be examined for the defects listed in table I. The lot size shall be expressed in units of primary containers. The sample unit shall be one filled primary container. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample data indicates that the lot average net weight is less than the specified net weight.

TABLE I. Net weight defects 1/

Category	Defect
Minor	
201	Net weight more than 5 percent under specified net weight for containers required to have 1 pound or less.
202	Net weight more than 2 percent under specified net weight for containers required to have more than 1 pound but less than 10 pounds.
203	Net weight more than 1 percent under specified net weight for containers required to have 10 pounds or more.

1/ Weigh to nearest 1/8 ounce for containers required to have 1 pound or less, nearest 1/2 ounce for containers required to have more than 1 pound but less than 10 pounds, and nearest 1 ounce for containers required to have 50 pounds or more.

4.5.4 Product inspection. The bakery mix and the baking soda, yeast, and albumen mix (when unit packed in separate pouches) shall be examined for the defects specified in table II. The sample unit shall be one filled and closed primary container. The lot size shall be expressed in units of primary containers. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

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TABLE II. Products defects 1/

Category		Defect
Major	Minor	
101		Pouch missing (when applicable)
102		Pouch tear, hole, or open seal (when applicable)
103		Pouch not clean (when applicable)
104		Pouch not labeled as specified (when applicable)
105		Pouch not fabricated as specified in 5.1 (when applicable)
	201	Mix not well blended or not uniform color
106		Foreign odor
107		Lumps which do not disintegrate upon finger pressure
	202	Pouch not placed as specified (when applicable)
108		Pouch leaks when examined as specified in 4.6.4 (when applicable)
109		Quantity of bicarbonate of soda/sodium bicarbonate more than 10 percent below or more than 20 percent above (the specific formulation requirement) by weight 2/
110		Evidence of delamination and/or seal separation when examined as specified in 4.6.4

1/ Presence of foreign material (e.g., insects, filth, metal, dirt, wood, glass, paint) shall be cause for rejection of the lot.

2/ Weigh to the nearest 0.1 gram and calculate the percent deviation from the required weight.

4.5.5 Moisture testing. The finished product (exclusive of leavening or egg whites pouches) shall be tested in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Sugar and Sugar Products, Section: Sugars and Sirups, Method: Vacuum Drying, modified for conformance to the moisture requirement of the applicable product. The sample for testing shall be a 1-pound composite of product derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of primary containers. Test results shall be reported to the nearest 0.1 percent. A test failure shall be cause for rejection of the lot.

4.5.6 Oxygen content of yeast pouch testing. The filled and sealed yeast pouch shall be tested for oxygen content in accordance with 4.6.5. The sample unit shall be one filled and sealed pouch. The sample size shall be the number of sample units indicated by inspection level S-2. Lot size shall be expressed in pouches. The AQL, expressed in terms of percent defective, shall be 2.5. Results shall be reported to the nearest 0.1 percent. A defect shall be scored for a test result greater than 3.0 percent.

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4.5.7 Cooked product examination. The mix shall be prepared to form a ready-to-eat item in accordance with the manufacturer's instructions except as otherwise directed in table III. The cooked product shall be examined for the applicable defects specified in table IV. The sample unit shall be one filled and closed primary container, except that for bulk packs the sample unit shall be 5 pounds extracted from a sack of mix. The lot size shall be expressed in units of primary containers. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 10.0 for minor defects.

TABLE III. Preparation of cooked product

Class	Style	Requirement paragraph Type I	Type II	Preparation exception paragraph
1	A	3.3.1.1.1	3.3.2.1.1	4.6.1.1
1	B	3.3.1.1.2	3.3.2.1.2	4.6.1.1
1	C	3.3.1.1.3	3.3.2.1.3	4.6.1.1
1	D	3.3.1.1.4	3.3.2.1.4	4.6.1.1
1	E	3.3.1.1.5		
1	F	3.3.1.1.6		
1	G	3.3.1.1.7	-	-
1	H	3.3.1.1.8	-	-
1	I	3.3.1.1.9	-	-
1	J	3.3.1.1.10	-	-
1	K	3.3.1.1.11	-	-
1	L	3.3.1.1.12	-	-
2	A	3.3.1.2.1	3.3.2.2.1	4.6.1.2
2	B	3.3.1.2.2	3.3.2.2.2	4.6.1.3
2	C	3.3.1.2.3	3.3.2.2.3	4.6.1.4
2	D	3.3.1.2.4		
2	E	3.3.1.2.5		
3	-	3.3.1.3	3.3.2.3	4.6.1.5
4	A	3.3.1.4.1	3.3.2.4.1	4.6.1.6
4	B	3.3.1.4.2	3.3.2.4.2	4.6.1.7
5	A	3.3.1.5.1	3.3.2.5.1	4.6.1.8
5	B	3.3.1.5.2	3.3.2.5.2	4.6.1.8
6	A	3.3.1.6.1	-	-
6	B	3.3.1.6.2	-	-

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TABLE IV. Cooked product defects 1/ 2/ 3/ 4/

Category		Defect
Major	Minor	
101		Product (type I, class 1, all styles except pound cake and angel food cake and type II, class 1, styles A, B, C and D) breaks when evaluated in accordance with 4.6.3
102		Lumps that do not dissolve in the baking process
	201	Product not symmetrical or nearly symmetrical, as required
	202	Color of crust or surface of product not as specified
	203	Contour of top of product not as required
	204	Doughnut hole not small and puckered
	205	Specific volume or volume of product not as specified
	206	Height of product not as specified 5/
	207	Condition of crust or surface of product not as specified
	208	Cake does not separate easily from pan or liner (e.g, sticks, cracks, breaks, or crumbles)
	209	Product breaks, cracks, or crumbles when handled
	210	Breaks or cracks (major or otherwise as specified) in product
	211	Product cannot be easily sliced
	212	Color or structure of crumb or grain not as specified
	213	Texture of crumb not as specified
	214	Large holes, tunnels, or gas pockets in product
	215	Corn bread or muffin cannot easily be broken into large distinct intact pieces by hand
	216	Top and bottom of biscuit not easily separated by hand
	217	Grease penetration of doughnut more than 1/16 inch in depth 5/
	218	Batter setting sample (type II, class 1, styles A, B and C only) exhibits crusty foam like top after baking
	219	Flavor or odor not typical of product as required by formulation, but not objectionable
	220	Product (type II, class 2, style C only) does not conform to specified dimensions

1/ The presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

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3/ Examine one cake of class 1 cake mixes, all styles; one loaf of class 2, style A, yellow corn bread mix, and class 4, styles A and B; and three units of class 2, style B, biscuit mix, class 2, style C pancake and waffle, regular class 3, cake doughnut mix, and class 5, styles A and B, chocolate and butterscotch brownie mix.

4/ No defect shall be scored more than once per sample unit.

5/ Measuring device shall be calibrated in 1/16-inch increments.

4.5.8 pH testing of baked cake. Type II, class 1, styles C and D baked cake shall be tested for pH in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Cereal Foods, Section: Wheat Flour, Method: Hydrogen Ion Concentration Potentrometric Method. The sample for testing shall be an 8-ounce composite derived from the baked product samples specified in 4.5.7. Results shall be reported to the nearest 0.1. A test failure shall be cause for rejection of the lot.

4.5.9 Examination of paper pockets and cartons. Filled and sealed paper pockets and cartons shall be examined for the defects specified in table V. The sample unit shall be one filled and sealed paper pocket or carton, as applicable. The lot size shall be expressed in units of paper pockets or cartons, as applicable. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for minor defects.

TABLE V. Paper pockets and carton defects 1/

Category		Defect
Major	Minor	
101		Open seam or seal, tear, hole, or sifter
102		Not clean
103		Nomenclature or instructions for use missing, incorrect, or illegible
	201	Other required label information missing, incorrect, or illegible

1/ Level B paper pockets without grease resistant liner or inner ply shall not be acceptable.

4.5.10 Can condition examination. Examination of filled and sealed cans shall be in accordance with the United States Standards for Condition of Food Containers and PPP-C-29, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

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4.5.11 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number or date of pack markings missing, incorrect, or illegible. Reinforced with other than nonmetallic strapping or tape.

Minor: Other required markings missing, incorrect, or illegible. Arrangement or number of cans not as specified.

Level C shipping containers shall be examined for the marking defects specified above and for the closure method specified in 5.2.2.

4.5.12 Shipping sack examination. When shipping sacks are required to be in accordance with UU-S-48, examination of the filled and closed sack shall be in accordance with the applicable quality assurance provisions of UU-S-48. Marking defects shall be classified as specified in 4.5.11.

4.5.13 Waterproof case liner examination. When applicable, waterproof case liners shall be examined in accordance with the quality assurance provisions of MIL-L-10547.

4.5.14 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

4.6 Methods of inspection.

4.6.1 Testing procedures for type II products. Batter shall be prepared according to directions specified in 5.4.1.2. In addition, the following requirements shall apply:

Type of cake mix	Amount of mix other than that in pouch to be used	Amount of baking soda to be used 1/	Amount of other ingredients to be used
White cake	2255g	11.3g	--
Yellow cake	2255g	12.5g	--
Devil's food cake	2240g	26.1g	--
Gingerbread	2245g	22.7g	--
Yellow corn bread	2030g	10.2g	27.6g sodium aluminum phosphate
Biscuit	1990g	27.6g	--
Pancake and waffle	1945g	39.7g	--
Doughnut, cake	2020g	19.2g	--
Brownie, chocolate	2260g	6.8g	--
Brownie, butterscotch	2260g	5.7g	--

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Mixer speed where specified shall be defined as follows:

Low speed is first speed on a 3-speed mixer and second speed on a 4-speed mixer.
Medium speed is second speed on a 3-speed mixer and third speed on a 4-speed mixer.

The mixer paddle shall be the attachment known as a flat agitator.

1/ At high altitude test sites, use proportionately less soda (i.e. at 2000 feet use 6/7 times amount listed above; at 4000 feet use 5/7 amount listed above, etc.)

4.6.1.1 Type II, Class 1, Styles A through D. Lightly grease the sides of an 8-inch round pan with 1-1/4 inch sides (an aluminum pan No. 5345 Wear-Ever or equivalent is acceptable) and add a release type parchment paper to the bottom of the pan. Add 450 grams of batter and bake at 350°F for 25 to 30 minutes until cake is done for styles A, B and C, and at 350°F for 30 to 40 minutes until cake is done for style D. Place the baked cake in its pan on a wire rack in a well-ventilated area and allow to cool at a room temperature of 74°F ± 6°F and a relative humidity of 50 ± 5 percent. Remove cake from the pan, remove paper, and determine its specific volume and run a break test (see 4.6.2 and 4.6.3). Slice cake in half vertically and examine its grain and texture. In addition for type II, class 1, styles A, B and C, place a second 450-gram sample of batter in a round pan as noted above, allow to set for 30 minutes at a temperature of 78° to 84°F, bake and cool as described above. Examine for conformance to the batter setting requirement.

4.6.1.2 Class 2, style A. Procedure shall be as follows: into an 8-inch round pan with 1-1/4 inch sides and lined on the bottom with a parchment disc, shall be scaled 672 grams of corn bread batter. Bake at 400° to 425°F for 20 to 30 minutes for the corn bread. Remove from oven and allow to cool at a room temperature of 74°F ± 6°F and a relative humidity of 50 ± 5 percent on a wire rack. When cool remove from the pan and examine contour. Determine specific volume in accordance with 4.6.2. Slice loaf in half vertically and horizontally. Examine for crumb, grain, and texture.

4.6.1.3 Class 2, style B. Procedure shall be as follows: turn out the dough between 1/2-inch gauge sticks on a lightly floured rolling board. Roll until the rolling pin rests on the sticks. Fold double and reroll at right angle to this fold. Cut out biscuits from the central portion of the dough and place them on a lightly greased cookie sheet. Biscuits must be at least 1/4 inch from each other while baking. Place in an oven at 425°F and bake for 15 minutes, or until the crust is a light brown. Remove from the oven and allow to cool at a room temperature of 74°F ± 6°F and a relative humidity of 50 ± 5 percent. Weigh three biscuits to the nearest gram and determine specific volume in accordance with 4.6.2.

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4.6.1.4 Class 2, style C. Utilizing the entire contents of the container, prepare batter in accordance with directions on container; allow batter to stand 10 minutes. Using a number 20 ice cream scoop, dip one scoop (2 ounces) liquid batter onto a lightly greased 375° to 385°F griddle. Fry for 75 seconds on first side, then turn and fry for 60 seconds.

4.6.1.5 Class 3. Cake doughnut mix shall be examined by the following procedure: prepare the mix according to the manufacturer's directions. Drop the doughnuts into frying fat having a temperature of 380°F. Fry until light brown in color on the underside. Flip the doughnuts over and continue frying until they are uniformly colored. Remove the doughnuts from frying fat and allow to cool at a room temperature of 74°F ± 6°F and a relative humidity of 50 ± 5 percent on a wire rack. Weigh 8 to 10 doughnuts to the nearest gram and determine the specific volume in accordance with 4.6.2. Slice a representative doughnut horizontally and observe the crumb.

4.6.1.6 Class 4, style A. Procedure shall be as follows: weigh 1000 grams of prepared mix and transfer to a 3-quart mixing bowl. Add 1 ounce of active dry yeast, suspended in the quantity of water which has been predetermined to give the optimum dough characteristics. The water temperature shall be so adjusted that the mixed dough will have a temperature of 81° ± 1°F. The dough shall be mixed with a dough hook until dough entirely cleans bowl, and for 1 additional minute thereafter. Remove the dough from the mixer and scale two pieces weighing 23 ounces each. Place each piece in a greased glass jar of approximately 5 inches inside diameter by 7 inches inside height and calibrated in mL. Cover jars with a loose-fitting lid. Place the jars in a fermentation cabinet set at 81° ± 1°F and a relative humidity of 75 to 85 percent. Ferment in the jars until the outer edge of the dough piece reaches the 1400 mL mark. Remove the jars from cabinet and punch and round dough to expel gas and air pockets. Return rounded pieces of dough to jars and place jars in fermentation cabinet for 15 minutes. After 15 minutes remove dough and mold each piece into a loaf shape. Place loaves in lightly greased 1-pound loaf bread pans and place in proof cabinet maintained at 96° to 100°F and 75 to 85 percent relative humidity. Proof for approximately 1 hour. After proofing, bake the loaves in a 425° ± 10°F oven for 30 minutes or until done. Remove loaves from the pans and allow to cool at a room temperature of 74°F ± 6°F and a relative humidity of 50 ± 5 percent on a wire rack. Determine the loaf volume in accordance with 4.6.2. Slice a loaf vertically through the center with a sharp knife and observe the internal characteristics.

4.6.1.7 Class 4, style B. Procedure shall be as follows: weigh 1000 grams of prepared mix and transfer to a 3-quart mixing bowl. Add 3/4 ounce of active dry yeast, suspended in the quantity of water predetermined to give the optium dough characteristics. The water temperature shall be so adjusted that the mixed dough will have a temperature of 81° ± 1°F. The dough shall be mixed with a dough hook until dough completely cleans bowl, and for 1 additional minute thereafter. When mixed, place dough in a greased jar or can and cover with a loose-fitting lid. Place dough in fermentation cabinet maintained at 81° ± 1°F

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with relative humidity of 75 to 85 percent. Allow dough to ferment 1-1/4 hours. Remove dough and scale two 22-ounce pieces. Round scaled pieces and allow to rest 15 minutes on bench, covering pieces with a cloth while resting. Mold into load shape and place in lightly greased 1-pound loaf pans. Proof loaves in a cabinet maintained at 96°F to 100°F and 75 to 85 percent relative humidity. Proof for approximately 1 hour. After proofing, bake the loaves in a 400°F ± 10°F oven until done (about 30 minutes). Remove loaves from the pans and allow to cool at a room temperature of 74°F ± 6°F and a relative humidity of 50 ± 5 percent on a wire rack. Determine the loaf volume in accordance with 4.6.2. Slice a loaf vertically through the center with a sharp knife and observe the internal characteristics.

4.6.1.8 Class 5, styles A and B. Bake 500 grams of batter at 350°F until done (35 to 40 minutes) in an 8- by 8-inch pan. The baking pan shall be lightly greased at the bottom. Place baked product in pan in a well-ventilated area and allow to cool to room temperature.

4.6.2 Specific or loaf volume. Determination of specific or loaf volume shall be accomplished by a loaf volumeter (8-inch round cake size) or equivalent (see 6.4) or by the rapeseed method (see below). Avoid any vibrations that would tend to pack the rapeseed.

(a) Remove parchment and place cooled product in a container.

(b) Fill the void space in the container containing the product with rapeseed so that the rapeseed is level with the top edge of the container. Measure this amount of rapeseed in whole milliliters, using a graduated cylinder or equivalent measuring device.

(c) Remove product from the container and weigh the product. The weight shall be calculated in terms of whole grams.

(d) Fill the empty container with rapeseed so that the rapeseed is level with the top edge of the container. Measure the amount of rapeseed as specified in (b) above. Alternatively, container volume may be established in the above manner, prior to putting baked or fried product (i.e., doughnuts) in a container.

(e) Calculate specific volume as follows:

$$\frac{\text{Volume of rapeseed (d)} - \text{Volume of rapeseed (b)}}{\text{Weight of product (c)}} = \text{Specific volume}$$

$$\text{Volume of rapeseed (d)} - \text{Volume of rapeseed (b)} = \text{Loaf volume}$$

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4.6.3 Breaks and cracks. Bake as in 4.6.1.

(a) Cool the cake in the pan for 2 hours at room temperature 74° ± 6°F and a relative humidity of 50 ± 5 percent.

(b) Space two 4-inch diameter cylinders (i.e., two flat topped metal cans) 6-1/2 inches apart at the closest point.

(c) Place the cake (with pan and liner removed) with the flat side down equally on the two cylinders.

(d) Examine after 2 minutes for breaks and cracks.

4.6.4 Leakage. The filled and sealed pouch shall be tested by placing in a dry desiccator, or similar apparatus, and subjecting to a vacuum of 26 inches of mercury (atmospheric pressure is 29.9 inches of mercury) for 30 seconds. Any pouch that does not swell to form a tightly distended package having at least one distorted edge during the test shall be recorded as a leaker. After leakage testing, the pouch shall be visually inspected for evidence of delamination and for seal separation. Any leakage, any delamination or any seal separation of more than 1/16 inch from the product edge of any seal shall be recorded as a defect.

4.6.5 Oxygen content test. Oxygen content of pouches shall be determined in accordance with the American Dry Milk Institute, Inc., publication, Bulletin 916, Standards for Grades of Dry Milks, Including Methods of Analysis: Determination of Oxygen. Alternatively, the oxygen content may be determined by an Instrumentation Laboratories Oxygen Analyzer or equivalent instrument. Test results shall be reported to the nearest 0.1 percent for each pouch. Any pouch having an oxygen content greater than 3.0 percent shall be considered a test failure. When referee testing is necessary, the American Dry Milk Institute method shall be followed.

5. PACKAGING

5.1 Preservation. The product shall be preserved level A, B, or C as specified (see 6.1). For type I, class 4 product, the required amount of yeast may be divided into no more than three equal portions and each shall be unit packed in a foil-laminated pouch that has been gas flushed and sealed to maintain an oxygen content of not more than 3.0 percent when tested in accordance with 4.6.5. For type II product, the required amount of bicarbonate of soda, dried egg whites with other ingredients, or sodium aluminum phosphate, as applicable, shall be separately unit packed in a foil-laminated pouch. The pouch shall be hermetically sealed and shall not leak when tested in accordance with 4.6.4. The pouches of modified egg whites with other ingredients or bicarbonate of soda or sodium aluminum phosphate, as applicable, shall be placed either at the bottom of the can or on top of the product before final sealing of the product container. The weight of the separately packed ingredients shall be included in the net weight for each unit pack.

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5.1.1 Level A or C cans (type II only). The following quantities of the product shall be unit packed into a size 603 by 700 open-top style, round, metal can in accordance with level A or C requirements of PPP-C-29 as specified.

Mixes	Weight
Class 1	
Styles A, B, C, D	5 pounds
Class 2	
Styles A and B	4 pounds 8 oz.
Style C	4 pounds
Class 3	4 pounds 8 oz.
Class 4	
Styles A and B	4 pounds 8 oz.
Class 5	
Styles A and B	5 pounds

5.1.2 Level B, paper pockets. Five pounds or the specified weight of the product shall be unit packed in commercial paper pockets constructed with an inner face or liner of grease-resistant material. The pocket shall be securely sealed to prevent sifting. For type I, class 4, style A or B, the net weight of active dry yeast shall be included in the net weight of the dry mix for each unit pack.

5.1.3 Level C. The type I product shall be preserved to afford adequate protection against deterioration and damage during shipment from the supply source to the first consuming activity. The contractor may use his standard practice when it meets these requirements.

5.2 Packing. The cans shall be packed in accordance with the level A, B, or C requirements of PPP-C-29 as specified (see 6.1). The product in cartons, paper pockets and bulk shall be packed in accordance with level B or C as specified (see 6.1).

5.2.1 Level B. The product shall be packed in accordance with 5.2.1.1, 5.2.1.2, or 5.2.1.3, as applicable.

5.2.1.1 Paper pockets. Ten 5-pound paper pockets or when other than 5 pounds per pocket is specified, then not more than 50 pounds of the product shall be packed in a multiwall paper shipping sack fabricated and closed in accordance with type I, style B; type II, style A or type VI, style B construction 4 or 4x, MB1, of UU-S-48.

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5.2.1.2 Bulk pack. Fifty pounds of bulk product shall be packed into a multiwall paper shipping sack as specified in 5.2.1.1.

5.2.1.3 Cartons. Not more than 50 pounds of the specified product in commercial cartons shall be packed into a snug-fitting fiberboard shipping container, constructed and closed in accordance with style RSC, V3c, V3s, or V4s of PPP-B-636. When specified (see 6.1), the cartons shall be overpacked into a waterproof case liner, fabricated and sealed in accordance with MIL-L-10547 for subsistence items, prior to packing the product inside the shipping container. Each shipping container shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament tape in accordance with the appendix of PPP-B-636.

5.2.2 Level C. Not more than 50 pounds of the specified product in commercial cartons, paper pockets, or bulk shall be packed in a shipping container in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable, except fiberboard shipping containers shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling and marking.

5.4.1 Unit containers. Cans shall be labeled in accordance with MIL-L-1497, and paper pockets and cartons shall be labeled in accordance with MIL-STD-129. Instructions for high altitude baking, and directions for use, as applicable, shall be in accordance with 5.4.1.1 and 5.4.1.2. For type II bakery mix in 50-pound sacks, instructions for baking shall be for 5, 4-1/2 and 4 pound increments of the mix, as applicable. Cans and paper pockets of type II product shall have the date of pack included in the labeling. Each can containing a pouch or pouches shall indicate on the can that the end with the pouch or pouches shall be opened first.

5.4.1.1 Labeling instructions. The following instructions for baking chemically leavened mixes at high altitudes shall be printed legibly on one side of the bicarbonate of soda pouch with ink which imparts no odor or taste to the baking mix; the printing shall be protected from smudging with a coating of lacquer or plastic film.

HIGH ALTITUDE BAKING INSTRUCTIONS

Lines and altitude markings on the reverse side of this pouch are to be used as a guide for high altitude baking, as follows:

1. Determine local altitude.

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2. Place pouch on a flat horizontal surface, pat gently to distribute contents evenly.

3. Holding pouch in horizontal position, cut across entire width of the pouch and contents with scissors or sharp knife along the line representing local altitude.

4. Discard portion of pouch with contents representing altitude below your present altitude.

5. Add contents of remaining portion of pouch to the mix and prepare according to "Directions for Use" on can.

6. Minor adjustments may be required on subsequent cakes.

The reverse side of the pouch shall be printed in accordance with the example shown below. The 2000 ft. line shall be printed across the width of the seal at 1/7 the distance from the inside of the top seal to the inside of the bottom seal. Subsequent 2000-foot altitude lines (e.g., 4000, 6000, and 8000 foot) shall be printed across the width of the seal printed at intervals of 2/7, 3/7 and 4/7 the distance between inside length dimensions.

(Example of required altitude marking)

LEAVENING AGENT (leavening acid or bicarbonate of soda as applicable)

At sea level or below 2,000 feet

2,000 ft. _____ USE ENTIRE CONTENTS _____

At altitudes above 2,000 ft. amount of leavening needed decreases

4,000 ft. _____

6,000 ft. _____

8,000 ft. _____

EXAMPLE: At 6,000 ft. cut along line marked 6,000 ft.
At 5,000 ft. cut midway between 4,000 ft. and 6,000 ft. lines.

5.4.1.2 Directions for use. The following directions for use as applicable shall be lithographed on the top of the can and printed on the paper pockets:

DIRECTIONS FOR USE

Type II, Class 1
Style A - White cake mix
Style B - Yellow cake mix
Style C - Devil's food cake mix

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5 lb Cake Mix Soda Pouch 4-3/4 cups (38 fluid ounces) water*

1. Place mix in mixer bowl. Add contents of soda pouch (at high altitudes use less soda - see pouch directions).
2. Blend with paddle at low speed** for 1 minute.
3. Add 1/3 of total water (approx 1-1/2 cups) immediately. Using paddle, mix in low speed** for 2 minutes.
4. Add remaining water gradually while mixing for 2 minutes at low speed**. Scrape bowl and paddle.
5. Mix in medium speed*** for 4 minutes.
6. Pour 3-3/4 quarts (7-1/4 lb) batter into a greased and floured sheet pan; spread evenly. For cupcakes: Fill each greased or paper-lined muffin tin with 1/4 cup (2 oz) batter (1 No. 16 scoop).

7. To Bake:	Conventional or rotary ovens****	Convection oven**** low fan, open vent
Sheetcakes	350°F 40 to 50 min.	300°F 25 to 35 min.
Cupcakes	350°F 20 to 30 min.	300°F 10 to 15 min.

8. For 100 portions for cakes (2 sheet pans) use 2 cans (10 lb), each portion: 1 piece (2 oz); for cupcakes 3-1/2 cans (17 lb 8 oz), each portion: 2 cupcakes (3-1/2 oz).

* Best results are obtained with a 70° to 75°F batter temperature.

** Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

*** Medium speed is second speed on a 3-speed mixer and third speed on a 4-speed mixer.

**** Bake time may require adjustment depending on the oven and oven load.

Type II, Class 1, style D - Gingerbread cake mix:

5 lb Cake Mix Soda Pouch 3-1/2 cups (28 fluid ounces) water*

15 lb. Cake Mix (2 sheet pans)

For 100 portions, use 3 cans (15 lb), each portion: 1 piece (2-3/4 oz).

1. Place mix in mixer bowl. Add contents of soda pouch (at high altitudes use less soda - see pouch directions).
2. Blend with paddle in low speed** for 1 minute.
3. Add 2/3 of total water (approx 1-3/4 qt per 100 portions) immediately. Using paddle, mix in low speed** for 2 minutes.
4. Add remaining water gradually while mixing for 2 minutes in low speed**. Scrape bowl and paddle.
5. Mix in medium speed*** for 4 minutes.
6. Pour 1 gallon (10-1/4 lb) batter into each of 2 greased and floured sheet pans; spread evenly.

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7. Bake at 350°F for 35 to 45 minutes. When a convection oven is used, bake at 300°F for 25 to 35 minutes on low fan, open vent.****

* Best results are obtained with a 70° to 75°F batter temperature.

** Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

*** Medium speed is second speed on a 3-speed mixer and third speed on a 4-speed mixer.

**** Bake time may require adjustment depending on the oven and oven load.

Type II, Class 2, style A - Yellow Corn Bread Mix:

4 lb 8 oz mix Soda Pouch 6-1/4 cups (50 fluid ounces) water*

Southern Style

1. Place mix in mixer bowl. Add contents of soda pouch (at high altitudes use less soda - see pouch directions).
2. Add water gradually while mixing for 1 minute in low speed** using paddle. Scrape bowl and paddle.
3. Mix in low speed** for 30 seconds.
4. Pour about 1 gallon (9 lb 8 oz) batter into each of 2 well-greased sheet pans or fill each greased muffin tin with 1/4 cup (2 oz) batter (1 No. 16 scoop).
5. Bake until done at 425°F 15 to 20 minutes for corn muffins, or 20 to 25 minutes for cornbread; for convection oven, bake at 375° on low fan, open vent, 15 minutes for corn muffins, or 20 minutes for cornbread.***
6. For 100 portions for cornbread (2 sheet pans) use 2-1/2 cans (11 lb 4 oz) each portion: 1 piece (2-1/2 oz); for corn muffins (17 12-cup muffin pans) use 4 cans (18 lb), each portion: 2 muffins (4-1/2 oz).

* Best results are obtained with a 70° to 75°F batter temperature.

** Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

*** Bake time may require adjustment depending on the oven and oven load.

Yankee Style

1. Add 6 oz (3/4 cup) sugar in step 1 (Southern style) for each can of mix while blending in soda.
2. Prepare as in Southern style.

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Type II, class 2, style B - Biscuit mix:

Biscuits

4 lb. 8 oz Mix	Soda Pouch and sodium aluminum phospate pouch	4-1/2 cups (36 fluid ounces) water*

1. Place mix in mixer bowl. Add contents of soda and sodium aluminum phosphate pouches (at high altitudes use less - see directions).
2. Blend with paddle in low speed** for 1 minute.
3. Add water gradually while mixing in low speed** for 30 seconds.
4. Fold over onto a lightly floured working surface. Knead lightly approximately 1 minute or until dough is smooth.
5. Roll out 1/2-inch thick and cut with a floured 2-1/2-inch biscuit cutter. Place biscuits on lightly greased sheet pan in rows 6 by 9.
6. Bake until lightly browned at 450° for 15 to 20 minutes; for convection oven, bake at 350°F for 15 minutes on low fan, open vent.***
7. For 100 portions (4 sheet pans), use 3-1/2 cans (15 lb 12 oz), each portion: 2 biscuits (3 oz).

* Best results are obtained with a 70° to 75°F batter temperature.

** Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

*** Bake time may require adjustment depending on the oven and oven load.

Pie Dough (Alternate Use)

1. Place dough mix in mixer bowl. (Discard both pouches. Do not add pouch contents to mix).
2. Add 7-1/2 ounces (1 cup) shortening and mix wilth paddle 1 minute on low speed*.
3. Add 2-3/4 cups cold water and mix with paddle 1 minute on low speed*.
4. Remove from bowl and shape into pie crust.
5. For 100 portions: 14 one-crust pies (1 can) or 14 two-crust pies (2 cans).

* Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

Type II, Class 2, Style C Pancake and waffle mix, regular:

4 lb. Mix (style C)	Soda Pouch	8-3/4 cups (70 ounces) water*

1. Place mix in mixer bowl. Add contents of soda pouch (at high altitudes use less soda - see pouch directions).
2. Blend with paddle in low speed** for 1 minute.
3. Add water gradually while mixing for 1 minute in low speed**.
4. Pour 1/4-cup batter on a lightly greased 375°F griddle. Cook on one side until top is covered with bubbles and underside is browned, turn and cook until other side is browned.

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5. For waffles, pour 1/2-cup batter on preheated lightly greased waffle iron. Bake until steaming stops.
6. For 100 portions, use 3-1/4 cans (13 lb).

* Best results are obtained with a 70° to 75°F batter temperature.

** Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

Type II, Class 3 - Cake Doughnut Mix

4 lb. 8 oz. Mix	Soda Pouch	3-3/4 to 4-1/2 cups (30 to 36 fluid ounces) water*

Machine Cut

1. Place mix in mixer bowl. Add contents of soda pouch (at high altitudes use less soda - see pouch directions). (Add 2 tbsp. ground nutmeg and 2 tbsp. vanilla per can).
2. Blend with paddle in low speed** for 1 minute.
3. Adjust water temperature so that batter temperature is 70° to 75°F.
4. Add water gradually while mixing in low speed** for 1 minute and medium speed*** for 2 minutes.
5. Let rest 10 minutes, then place in doughnut cutter.
6. Drop doughnuts in 375°F deep fat, fry 1 minute on each side or until lightly browned.
7. For 100 portions, use 4 cans (18 lb).

* Best results are obtained with a 70° to 75°F batter temperature.

** Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

*** Medium speed is second speed on a 3-speed mixer and third speed on a 4-speed mixer.

Hand Cut:

Follow mixing directions above, except use 3 to 3-3/4 cups (24 to 30 fluid ounces) of water. Place on well floured bench for make-up. (Allow about 8 oz. flour per can). Dough should be quite moist. Roll dough 3/8 inch thick, cut with doughnut cutter. Drop doughnuts in 375°F fat. Fry 1 minute on each side or until lightly browned.

Type II, Class 4, Style A - Roll Mix

4 lb. 8 oz. Roll Mix	(Yeast not included)	4-3/4 cups (38 fluid ounces) water

Rolls

1. Place 8 fluid ounces (105° to 110°F) water in mixer bowl (NOTE: DO NOT USE TEMPERATURES ABOVE 110°F).

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2. Dissolve 1-1/2 oz. active dry yeast in water, let stand 5-minutes. Add water which has been adjusted in temperature so that final dough temperature is 80°F.
3. Using dough hook, add remaining water while mixing 3 to 5 minutes at low speed* until water is absorbed.
4. Mix at medium speed** 3 to 5 minutes or until dough is developed and cleans the bowl.
5. Ferment at 80° to 86°F until double in bulk (about 1 hour).
6. Scale dough and make up in desired style. Place on lightly greased sheet pans.
7. Proof at 90° to 100°F until double in size (about 1 hour).
8. Bake until lightly browned in 400°F oven for 15 minutes or, for convection oven, bake at 325°F, 10 to 15 minutes on high fan, open vent.
9. For 100 portions, use 3 cans (13 lb 8 oz).

* Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

** Medium speed is second speed on a 3-speed mixer and third speed on a 4-speed mixer.

Pizza Dough

1. Follow Steps 1 thru 4 above.
2. Press dough not more than 1/4 inch thick on well oiled sheet pans. Bake at 400°F for 30 minutes or until slightly browned. For convection oven, bake at 350°F for 7 minutes on high fan, closed vent.
3. Add pizza sauce, mozzarella or pizza blend cheese and parmesan cheese. Add other toppings (i.e., green peppers, mushrooms, pepperoni and sausage) if desired.
4. Bake at 400°F for 30 minutes or until cheese starts to turn golden. For convection oven, bake at 350°F for 10 minutes on high fan, closed vent.
5. For 100 portions (4 sheet pans), use 1-1/2 cans (6 lb 12 oz).

Type II, Class 4, Style B - Sweet Dough

4 lb. 8 oz. Sweet Dough	(Yeast not included)	3-1/2 cups (28 fluid ounces) water

Sweet Rolls

1. Place water (105° to 110°F) in mixer bowl (NOTE: DO NOT USE TEMPERATURES ABOVE 110°F).
2. Sprinkle 1-1/2 oz. active dry yeast on top of water. Stir to dissolve. Let stand 5 minutes. Add mix.
3. Using dough hook, mix at low speed*, 3 to 5 minutes or until water is absorbed.
4. Mix at medium speed**, 3 to 5 minutes or until dough is developed and cleans the bowl.
5. Ferment at 80° to 86°F until double in size (about 1 hour).

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6. Scale dough and make up in desired styles. Dusting lightly with flour may be required. Place on greased sheet pans.
7. Proof at 90° to 100°F (45 to 60 minutes) or until double in size.
8. Bake until lightly browned in 400°F oven for 15 minutes or, for convection oven, bake at 325°F, 10 to 15 minutes on high fan, open vent.
9. For 100 portions, use 3-1/3 cans (15 lb).

* Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

** Medium speed is second speed on a 3-speed mixer and third speed on a 4-speed mixer.

Raised Doughnuts

1. Follow steps 1 thru 4 above.
2. Ferment at 80° to 86°F for 30 minutes.
3. Roll out 3/8-inch thick and cut with floured doughnut cutter. Place on greased and floured sheet pans.
4. Proof at 90° to 100°F for 30 minutes.
5. Fry at 370°F until golden brown on each side. Drain on absorbent paper.
6. For 100 portions, use 4 cans (18 lb).

Type II, Class 5, Style A - Chocolate Brownie Mix; Style B - Butterscotch Brownie Mix:

5 lb Mix	Soda Pouch	2 cups (16 fluid ounces) water* (chewy style) 3 cups (24 fluid ounces) water* (cake style)

1. Place mix in mixer bowl. Add contents of soda pouch (at high altitudes use less soda - see pouch directions).
2. Blend with paddle at low speed** for 1 minute.
3. Add water gradually while mixing 1 minute in low speed**. Scrape bowl and paddle. Mix in medium speed for 2 minutes***.
4. Add 4 fluid ounces (1 cup) chopped nuts for each can of mix (optional). Mix 30 seconds in low speed**.
5. Spread 1 gallon (9 lb) batter evenly into each greased sheet pan. For cake style: Spread 4-3/4 qt (9 lb 10 oz) batter evenly into each greased sheet pan.

6. To bake:

	350° conventional oven****	300° convection oven**** low fan, open vent
Chewy style	35 to 40 min.	30 to 35 min.
Cake style	30 to 35 min.	20 to 25 min.

7. For 100 portions (2 sheet pans), use 3 cans (15 lb).

* Best results are obtained with a 70° to 75°F batter temperature.

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** Low speed is first speed on a 3-speed mixer and second on a 4-speed mixer.

*** Medium speed is second speed on a 3-speed mixer and third speed on a 4-speed mixer.

**** Bake time may require adjustment depending on the oven and oven load.

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129 and UU-S-48, as applicable, and as specified in 5.4.3.1.

5.4.2.1 Additional marking for shipping containers. The following markings shall appear upon the top of each shipping case in letters not less than 1-inch high; for each shipping sack, the markings shall be located in any undesignated area in letters not less than 1-inch high.

STORE IN COOL, DRY PLACE
PRODUCT DETERIORATES RAPIDLY
ABOVE 80°F

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Types, classes, and styles of product desired (see 1.2.1).
c. When a first article is required (see 3.1, 4.4, and 6.3).
d. Provisions for approved preproduction samples (see 3.4.1 and 6.3).
e. Preservation and packing levels (see 5.1 and 5.2).
f. When cartons are to be waterproofed with a case-liner (see 5.2.1.3).
g. Type and class of unit load when unit loading is specified (see 5.3).

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item to be procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 7106/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all procurement documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Volumeter availability. A loaf volumeter manufactured by National Manufacturing Co., P.O. Box 30226, Lincoln, Nebraska, has been found to accurately test for volume.

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6.5 Subject term (key word) listing.

Bakery mixes
Biscuit mixes
Brownie mixes
Cake mixes
Doughnut mixes
Pancake mixes

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review Activities:

Army - MD, TS
Navy - MC, MS
DP - SS

Preparing Activity:

Army - GL

Project No. 8920-0479

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44275	Bakery Mixes, Prepared

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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